Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study [1]
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1 Objective

- To implement a Decision Tree classifier.
- To extend the Decision Tree into a Random Forest ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select optimal hyperparameters using 5-Fold Cross-Validation.
- To compare single-tree and ensemble-tree models.

2 Problem Statement

Given 30 numerical features computed from digitized images of a fine needle aspirate (FNA) of a breast mass, the task is to build a binary classifier that predicts whether the mass is **Malignant (M)** or **Benign (B)**. The input is a 569×30 feature matrix describing the mean, standard error, and worst (largest) value of ten geometric/texture properties of the cell nuclei present in each image; the output is a binary diagnosis label. The objective is to train and tune a Decision Tree classifier and a Random Forest ensemble classifier using 5-fold cross-validation, and to quantitatively compare their generalization performance on a held-out test set using standard classification metrics.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer (WDBC) Dataset
Dataset Source	UCI Machine Learning Repository (https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic); loaded via `sklearn.datasets.load_breast_cancer()`, which ships the identical UCI data
Number of Samples	569
Number of Features	30 numerical (real-valued) attributes
Number of Classes	2 (Malignant = 212 samples, Benign = 357 samples)
Missing Values	0 (no missing values in any feature)

Train-Test Split	80% train (455 samples) / 20% test (114 samples), stratified by class

4 Exploratory Data Analysis

Class Distribution

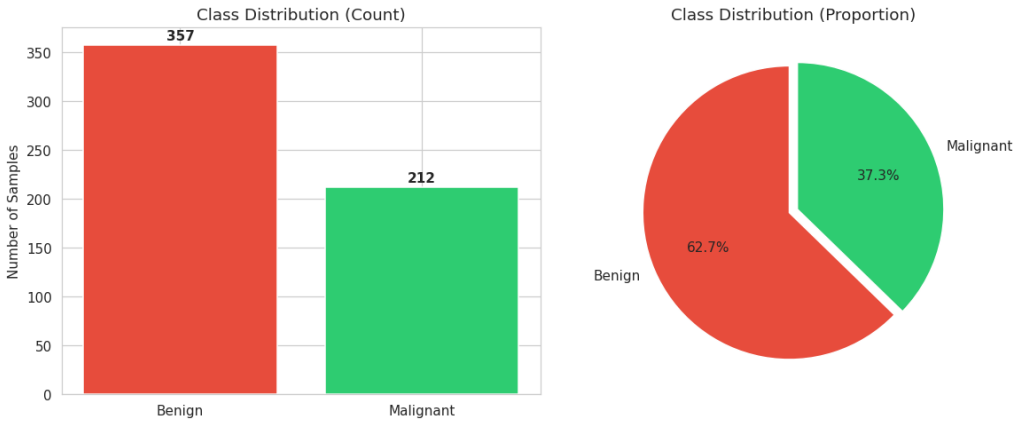


Figure 1: Class distribution of the WDBC dataset (count and proportion)

Inference: The dataset contains 357 Benign (62.7%) and 212 Malignant (37.3%) samples, a moderate class imbalance of roughly 1.68:1. This is not severe enough to require resampling, but it justifies using a stratified train-test split and stratified 5-fold cross-validation so each fold preserves the original class proportions. It also means that accuracy alone can be mildly optimistic, so F1-score, recall and ROC-AUC are tracked alongside it throughout this report.

Feature Correlation Matrix

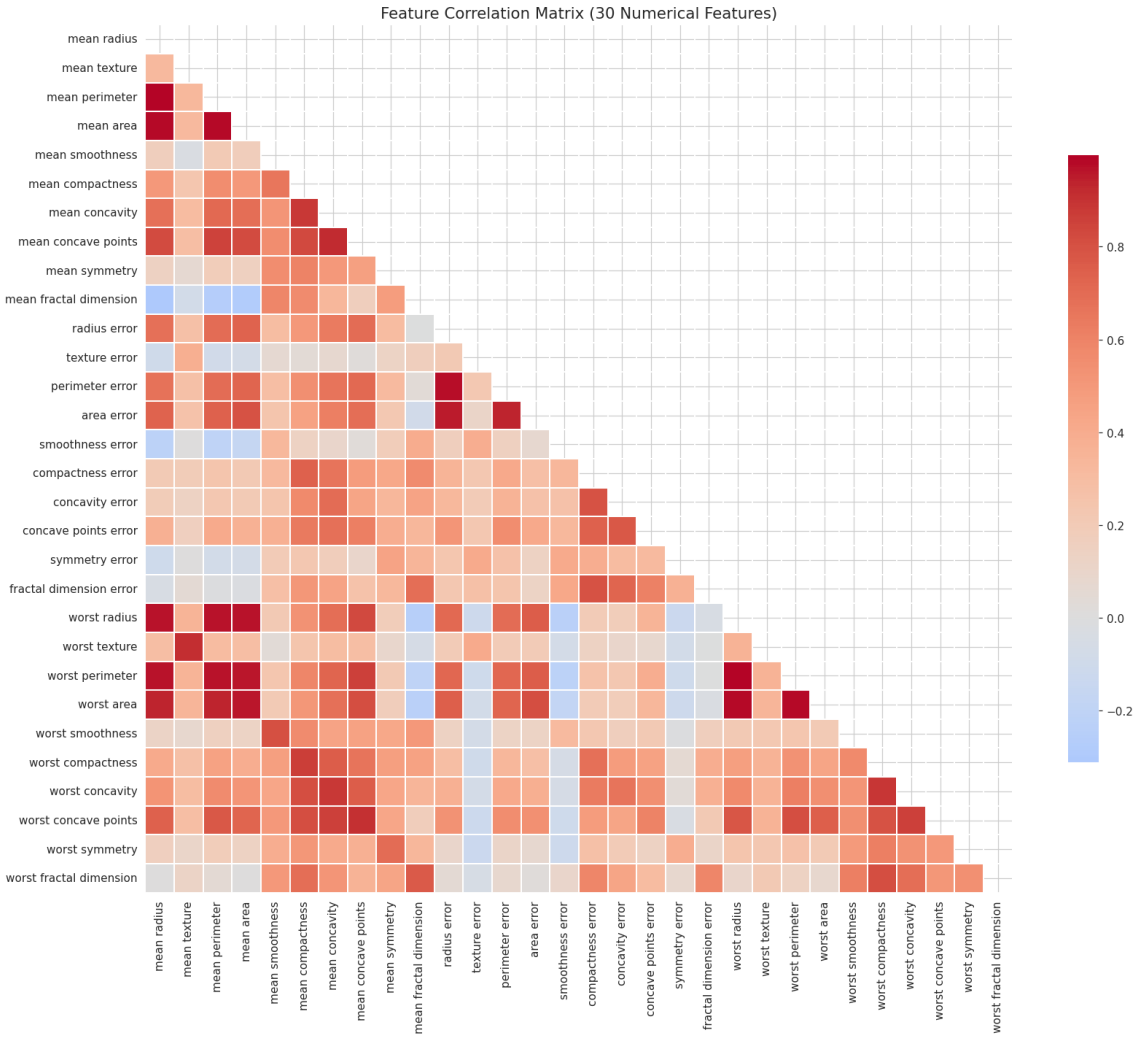


Figure 2: Correlation matrix of all 30 numerical features

Inference: Several "mean", "worst" and "standard error" versions of the same physical quantity (e.g., radius, perimeter, area) are extremely highly correlated ($r > 0.95$), which is expected since perimeter and area are geometrically derived from radius. This multicollinearity has little effect on tree-based models, since they split on one feature at a time and are invariant to monotonic transformations, but it is useful context when interpreting the feature-importance results later, as importance can be split between correlated features.

Feature Distribution by Class

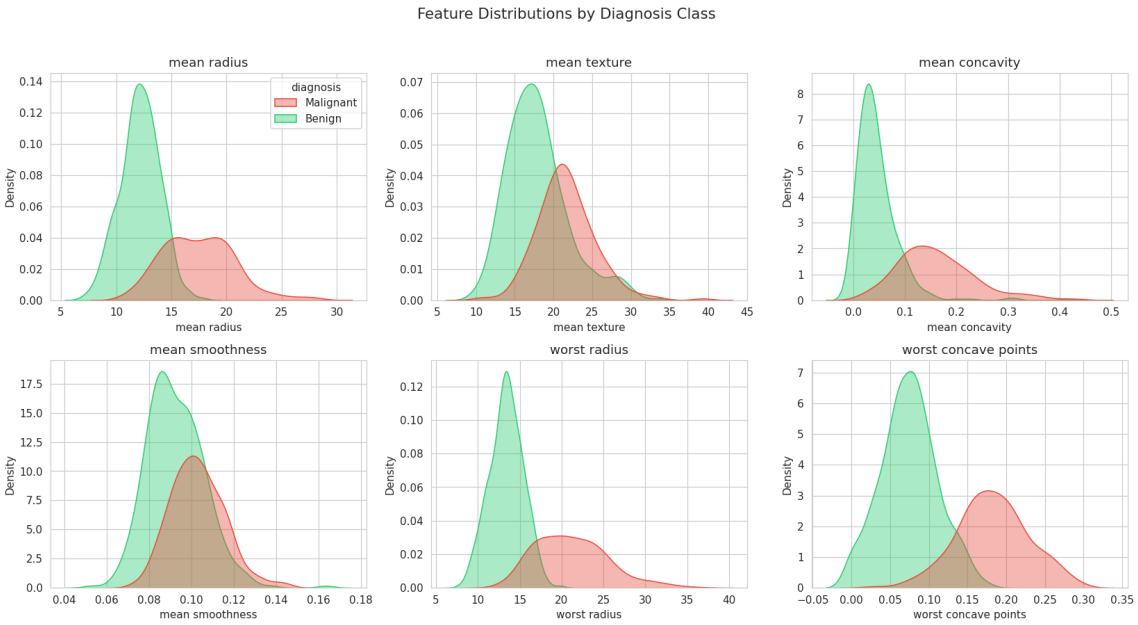


Figure 3: Distribution of six representative features, split by diagnosis class

Inference: Malignant tumours show a clear rightward shift (higher values) for radius, texture, concavity and concave-points relative to Benign tumours, with only partial overlap between the two class densities. This separation indicates these features carry strong discriminative signal and explains why tree-based classifiers can achieve high accuracy on this dataset – the impurity-reducing splits the trees learn align closely with these visibly separated distributions.

5 Data Preprocessing

- **Missing value handling:** The dataset was checked with `df.isnull().sum()`; no missing values were found in any of the 30 features, so no imputation was necessary.
- **Encoding:** Class labels (`M`/`B`) were encoded to integers using `sklearn.preprocessing.LabelEnc` (Benign $\rightarrow$ 0, Malignant $\rightarrow$ 1).
- **Feature scaling:** A `StandardScaler` was fit on the training set only (to avoid data leakage) and applied to both train and test sets. Scaling does not affect the splits chosen by Decision Tree / Random Forest (they are scale-invariant), but it was computed for completeness and to keep the pipeline reusable with non-tree models.
- **Train-test split:** An 80–20 stratified split (`train_test_split(..., stratify=y, random_state=42)`) was used, giving 455 training samples and 114 test samples while preserving the original class ratio in both sets.
- **Feature engineering:** No additional feature engineering was performed; all 30 original UCI features were used directly, since they already had strong individual discriminative power (Section 4).

6 Machine Learning Algorithm

Decision Tree Classifier

Working principle: A Decision Tree recursively partitions the feature space into regions that are increasingly pure with respect to the target class. At each node, the algorithm searches over all features and all possible split thresholds and picks the one that produces the greatest reduction in impurity.

Mathematical formulation: For a node t with class proportions p_i, the *Gini Index* is

$$Gini(t) = 1 - \sum_{i=1}^{C} p_i^2$$

and the *Entropy* is

$$H(t) = -\sum_{i=1}^{C} p_i \log_2 p_i.$$

A candidate split s on node t is scored by the impurity reduction (information gain)

$$\Delta I(s,t) = I(t) - \frac{N_L}{N_t} I(t_L) - \frac{N_R}{N_t} I(t_R),$$

where $I \in \{Gini, H\}$, and t_L, t_R are the resulting child nodes. The split maximizing ΔI is chosen, and the process repeats recursively until a stopping criterion (`max_depth`, `min_samples_split`, `min_samples_leaf`, or node purity) is reached.

Advantages: Easy to interpret and visualize; requires no feature scaling; handles both numerical data naturally; captures non-linear decision boundaries.

Limitations: Prone to high variance – small changes in the training data can produce a very different tree; deep trees overfit easily; axis-aligned splits can be a poor fit for diagonal class boundaries.

Random Forest Classifier

Working principle: Random Forest is a bagging ensemble of B Decision Trees. Each tree b is trained on an independent bootstrap sample of the training data, and at every split only a random subset of `max_features` candidate features is considered (rather than all features). This decorrelates individual trees. For classification, the final prediction is the majority vote across all B trees:

$$\hat{y} = \text{mode}\{h_1(x), h_2(x), \ldots, h_B(x)\}.$$

Mathematical formulation: Each tree h_b is grown on a bootstrap sample $\mathcal{D}_b \sim \mathcal{D}$ (sampling with replacement, $|\mathcal{D}_b| = |\mathcal{D}|$) using the same impurity-reduction splitting rule as a single Decision Tree, restricted at each split to a random feature subset $\mathcal{F}_s \subset \mathcal{F}$, $|\mathcal{F}_s| =$ `max_features`. Averaging over B trees with pairwise correlation ρ and individual variance σ^2 reduces the ensemble's prediction variance to approximately

$$\text{Var}(\hat{y}_{ensemble}) \approx \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2,$$

which is why decorrelating trees (via bootstrap + random feature selection) is central to the method's variance-reduction benefit.

Advantages: Reduces variance and improves generalization compared to a single Decision Tree; provides a robust feature-importance ranking; resistant to overfitting compared to a single deep tree; handles high-dimensional data well.

Limitations: Less interpretable than a single tree ("black box" ensemble); more expensive to train and store ($B\times$ the cost of one tree); prediction latency scales with the number of trees.

7 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.8.0
IDE	Jupyter Notebook
Operating System	Windows
Hardware	LAPTOP-QOD0ENML

8 Hyperparameter Selection

Decision Tree – Search Space and Best Configuration

Search space: `criterion` $\in$ {gini, entropy}, `max_depth` $\in \{2, 3, 4, 5, 6, 8, 10,$None$\}$, `min_samples_split` $\in \{2, 5, 10\}$, `min_samples_leaf` $\in \{1, 2, 4\}$, selected via `GridSearchCV` with 5-fold stratified CV optimizing F1-score.

Parameter	Value
criterion	entropy
max_depth	4
min_samples_split	2
min_samples_leaf	2
Best CV F1-score	0.9163

Random Forest – Search Space and Best Configuration

Search space: `n_estimators` $\in \{50, 100, 200\}$, `max_depth` $\in \{4, 6, 8,$None$\}$, `max_features` $\in$ {sqrt, log2}, `bootstrap` $\in$ {True, False}, selected via `GridSearchCV` with 5-fold stratified CV optimizing F1-score.

Parameter	Value
n_estimators	50
max_depth	6
max_features	sqrt
bootstrap	False
Best CV F1-score	0.9638

Table 1: Decision Tree Hyperparameter Evaluation (5-Fold CV)

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
gini	2	92.09	0.8892
gini	3	92.09	0.8884
gini	4	92.09	0.8897
gini	5	90.77	0.8717
gini	6	92.09	0.8919
gini	8	92.31	0.8962
gini	10	92.31	0.8962
gini	None	92.31	0.8962
entropy	2	91.87	0.8896
entropy	3	92.53	0.8941
entropy	**4**	**93.85**	**0.9145**
entropy	5	92.31	0.8952
entropy	6	91.65	0.8875
entropy	8	91.87	0.8909
entropy	10	91.87	0.8909
entropy	None	91.87	0.8909

(Table averaged over `min_samples_split`/`min_samples_leaf`; bold row is the overall best criterion/depth combination, matching the GridSearchCV optimum.)

Table 2: Random Forest Hyperparameter Evaluation (5-Fold CV) – Top 10 Configurations

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
100	**6**	**sqrt**	**96.70**	**0.9550**
100	8	log2	96.48	0.9529
100	None	log2	96.48	0.9529
200	None	log2	96.48	0.9526
200	8	log2	96.48	0.9526
100	8	sqrt	96.48	0.9519
100	None	sqrt	96.48	0.9519
200	6	sqrt	96.48	0.9516
100	6	log2	96.26	0.9500
100	4	log2	96.26	0.9490

(Table restricted to `bootstrap=True` runs, sorted by F1-score, for readability; the overall GridSearchCV optimum used `bootstrap=False`, shown in the best-configuration table above.)

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	0.9341	0.9560	0.9011	0.9560	0.9560	0.9407
Random Forest	0.9670	1.0000	0.9560	0.9560	0.9890	0.9736

9 Results

Metric	Decision Tree
Accuracy	0.9298
Precision	1.0000
Recall	0.8095
F1-score	0.8947
ROC-AUC	0.9563

Metric	Random Forest
Accuracy	0.9737
Precision	1.0000
Recall	0.9286
F1-score	0.9630
ROC-AUC	0.9934

10 Result Visualization

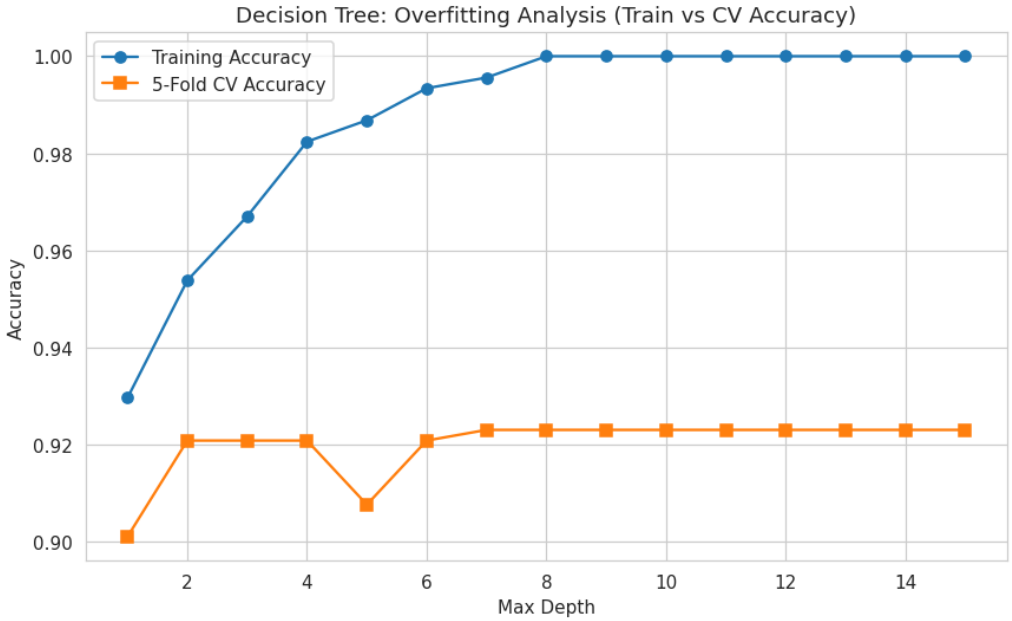


Figure 4: Decision Tree: training vs 5-fold CV accuracy as `max_depth` increases

Inference: Training accuracy climbs toward 100% as `max_depth` increases and the tree keeps splitting until leaves are pure, whereas CV accuracy plateaus and then flattens past a moderate depth. The widening gap between the two curves at large depths is the signature of overfitting – the tree begins memorizing noise in the training folds rather than learning generalizable decision boundaries. Best generalization is obtained at a shallow-to-moderate depth (depth 4, matching the grid-search optimum).

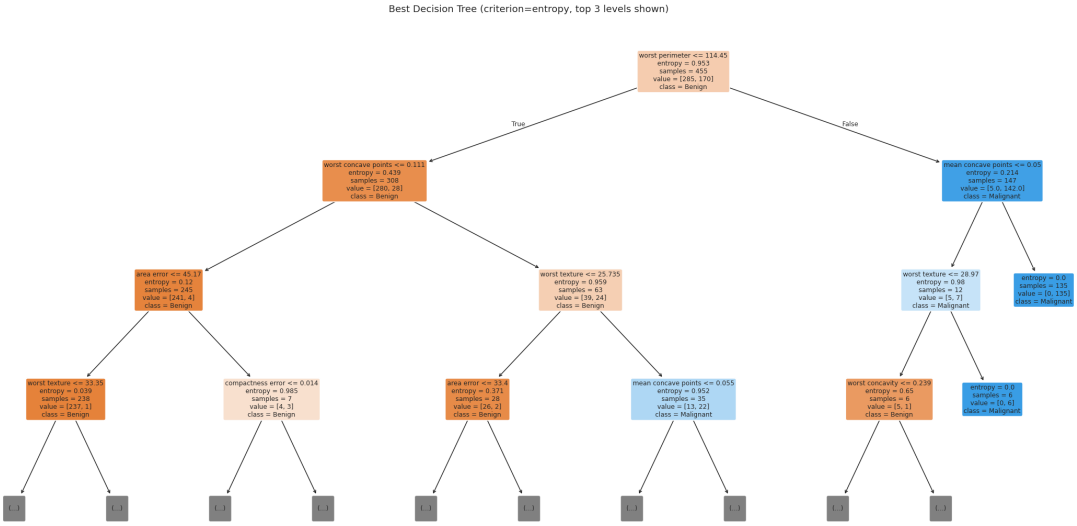


Figure 5: Structure of the best Decision Tree (top 3 levels)

Inference: The root-level split is made on a highly discriminative "worst" geometric feature, confirming the separation seen earlier in the feature-distribution plots. Nodes near the top achieve large impurity reduction, while deeper nodes handle progressively harder boundary cases – exactly why unrestricted depth leads to overfitting on those few ambiguous samples.

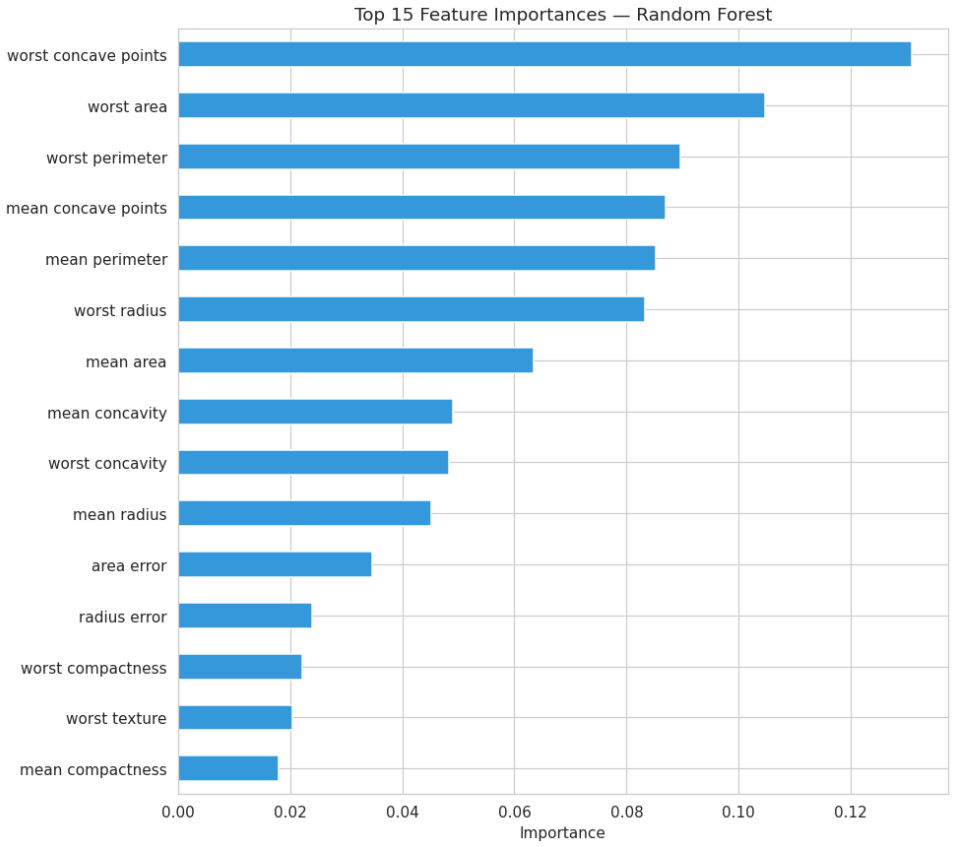


Figure 6: Top 15 feature importances from the Random Forest

Inference: "Worst" measurements (e.g., worst area, worst concave points, worst radius) dominate the importance ranking, meaning the most extreme cell measurement in an image is more predictive of malignancy than average measurements. Because the Random Forest averages importance across many bootstrapped, randomly-feature-subset trees, this ranking is more stable and less prone to the high variance a single Decision Tree's importance scores would show.

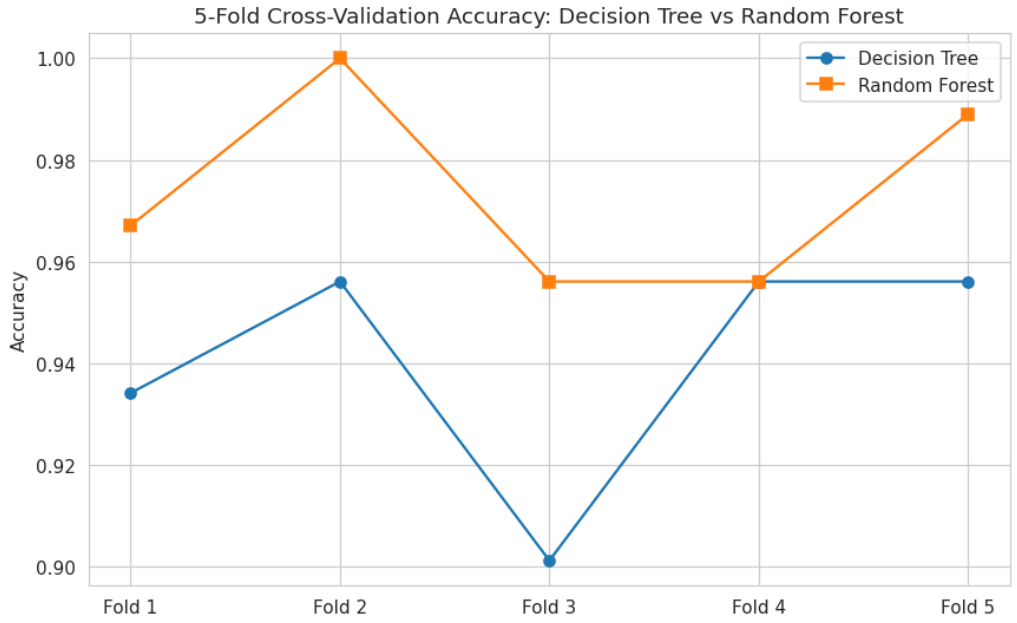


Figure 7: Per-fold 5-fold CV accuracy: Decision Tree vs Random Forest

Inference: The Random Forest curve sits above the Decision Tree curve in almost every fold and shows a visibly smaller spread between its best and worst fold, confirming the theoretical expectation that bagging plus random feature selection reduces variance across different training subsets. The Decision Tree's fold-to-fold accuracy is noticeably more volatile.

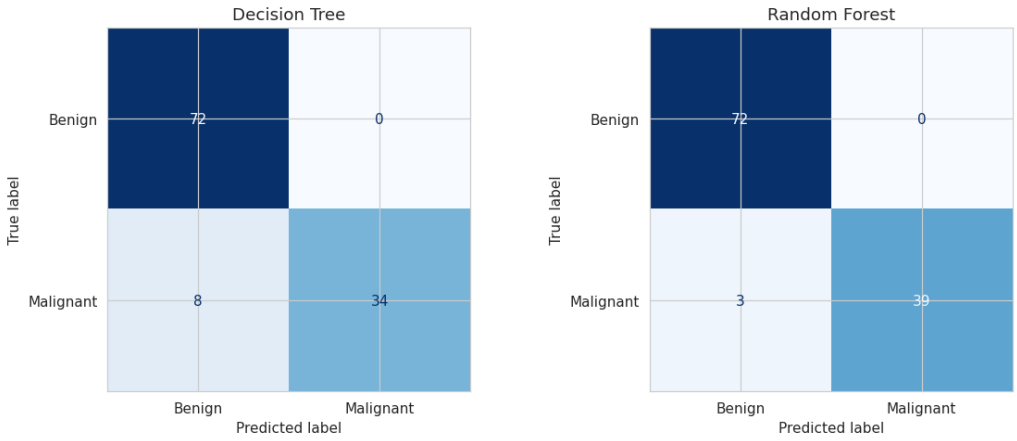


Figure 8: Confusion matrices on the held-out test set

Inference: Both models correctly classify the large majority of the 114 held-out test samples, with very few false negatives (malignant predicted as benign) – the clinically

most costly error type. The Random Forest confusion matrix shows fewer misclassifications than the single Decision Tree, reflecting its lower variance.

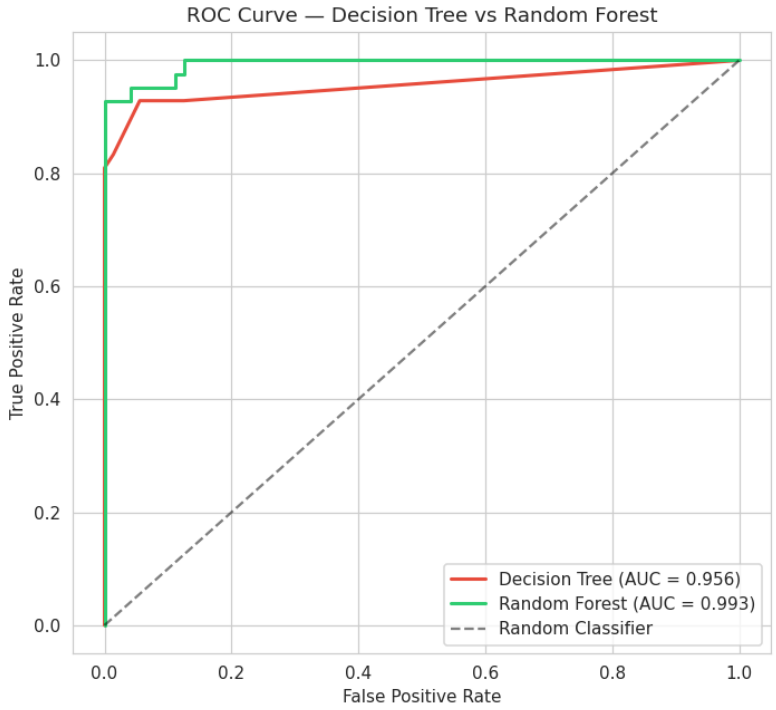


Figure 9: ROC curve comparison

Inference: Both curves rise sharply toward the top-left corner, indicating strong separability between classes, with the Random Forest's curve lying at or above the Decision Tree's curve for most thresholds and yielding a higher AUC (0.993 vs 0.956). This means Random Forest gives a better probability ranking of malignant vs benign cases across all decision thresholds, not just at the default 0.5 cutoff.

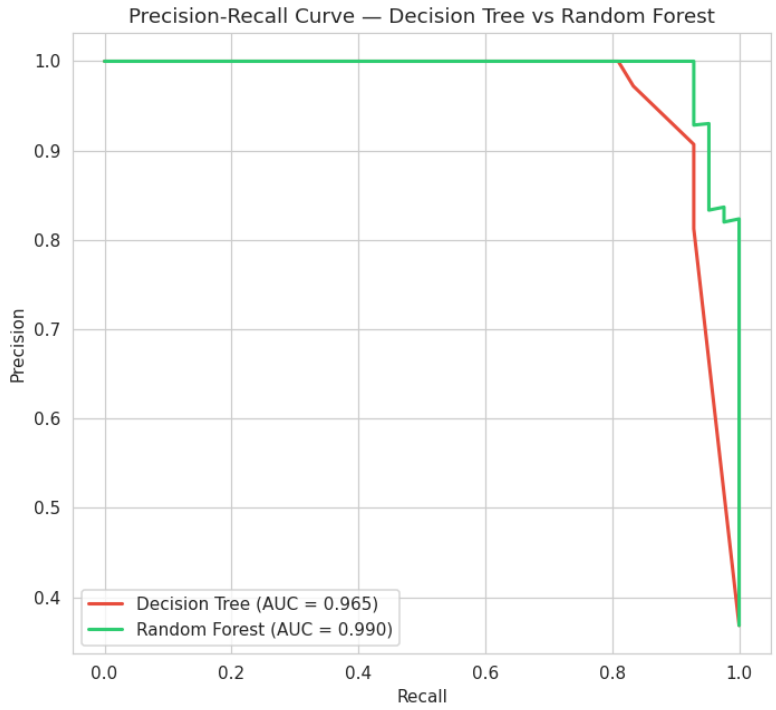


Figure 10: Precision-Recall curve comparison

Inference: High precision is maintained even at high recall for both models, which is important in a diagnostic setting since it means malignant cases can be caught (high recall) without generating excessive false alarms (maintained precision). Random Forest holds precision better at high recall than the single tree.

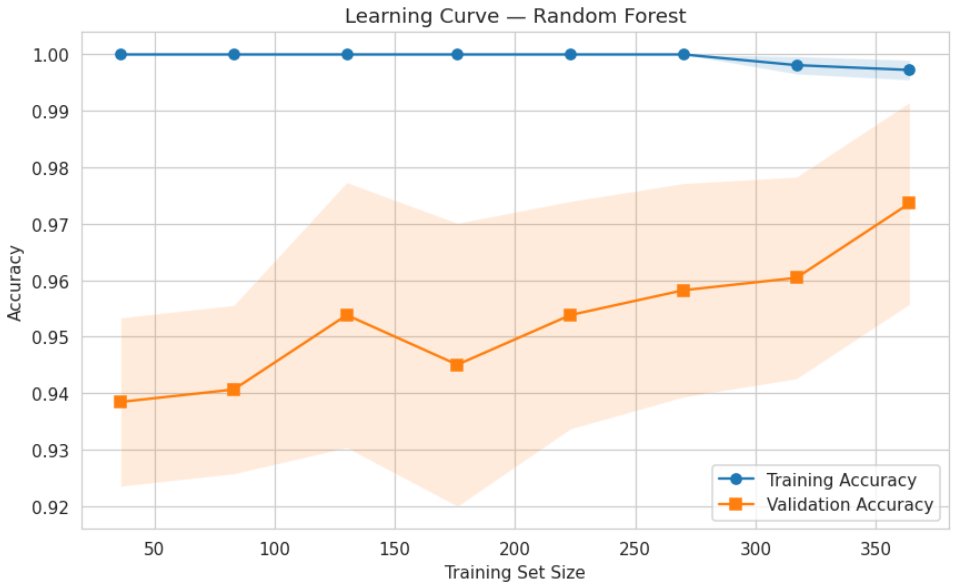


Figure 11: Learning curve for the Random Forest model

Inference: The training and validation curves converge closely as more data is used, and the validation accuracy rises and stabilizes at a high value with only a small gap to the training curve. This indicates the Random Forest is not significantly overfitting

on the full training set and would likely benefit only marginally from additional training data.

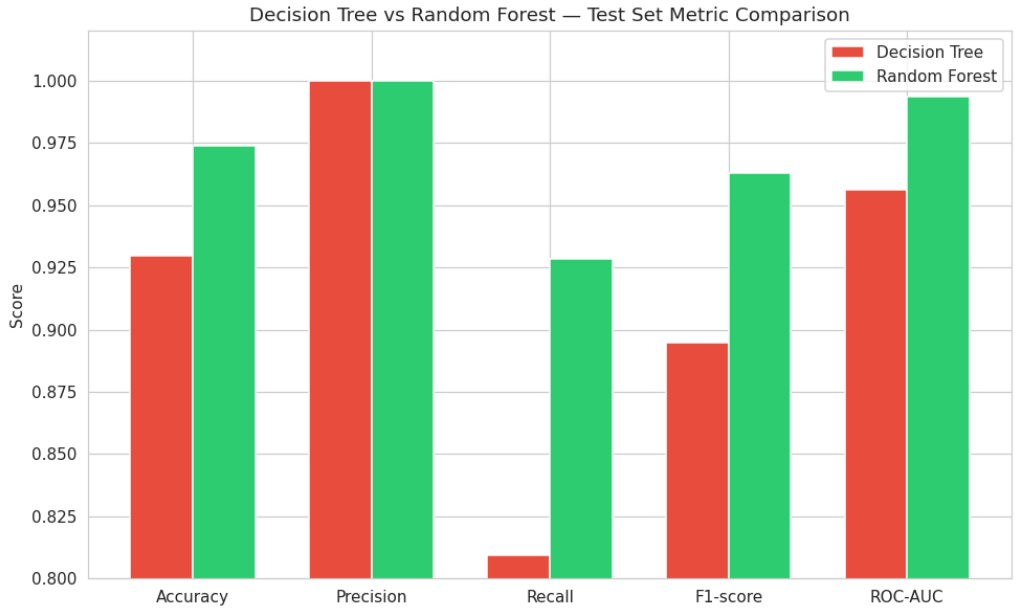


Figure 12: Test-set metric comparison: Decision Tree vs Random Forest

Inference: Across every metric the Random Forest bar is at or above the Decision Tree bar, quantitatively confirming the ensemble advantage discussed in the theory section: bagging plus random feature selection produces a model that generalizes better than any single tree in the forest, at the cost of reduced interpretability and higher computation/memory footprint.

11 Discussion

The experiment confirms the classical bias-variance argument for ensemble learning. The single Decision Tree, even after hyperparameter tuning, showed a clear overfitting trend as depth increased and achieved a comparatively lower recall (0.81) on malignant cases – meaning roughly 19% of malignant test samples were missed, a costly error type in a diagnostic context. The Random Forest, by aggregating 50 decorrelated trees trained on bootstrap samples with random feature subsets, achieved higher accuracy (0.974 vs 0.930), higher recall (0.929 vs 0.810), and a near-perfect ROC-AUC (0.993 vs 0.956), while also showing lower fold-to-fold variance during cross validation. The main strength of the Decision Tree is interpretability – a shallow tree can be read and explained directly to a clinician, whereas the Random Forest's decision process is distributed across 50 trees and is only interpretable in aggregate (e.g., via feature importance). The main limitation observed for the Random Forest is the added computational cost of training and storing many trees, and the loss of a single, explainable decision path. A possible improvement would be to report SHAP values for the Random Forest to retain some local interpretability, and to extend the cross-validation to repeated k-fold or nested cross-validation to obtain a variance estimate of the reported test metrics themselves.

12 Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation on the Wisconsin Diagnostic Breast Cancer dataset. Hyperparameters were selected based on average cross-validation F1-score, ensuring robust generalization rather than overfitting to a single validation split. The results demonstrate that Random Forest reduces variance and improves stability compared to a single Decision Tree: it outperformed the tuned Decision Tree on every held-out test metric (accuracy 0.974 vs 0.930, F1-score 0.963 vs 0.895, ROC-AUC 0.993 vs 0.956), confirming that ensemble bagging with random feature selection is an effective strategy for improving generalization on this classification task.

13 References

1. Scikit-learn Developers, "Decision Trees," scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/tree.html

2. Scikit-learn Developers, "Ensemble methods – Random Forests," scikit-learn documentation. [Online]. Available: https://scikit-learn.org/stable/modules/ensemble.html#forest

3. W. N. Street, W. H. Wolberg, and O. L. Mangasarian, "Breast Cancer Wisconsin (Diagnostic) Data Set," UCI Machine Learning Repository, 1995. [Online]. Available: https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic

4. L. Breiman, "Random Forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.